Intersection of 2 Lines

1. Find the vector equation of a line passing through $(3,0,2)$ and:

a) perpendicular to both $l_1 : (x,y,z) = (1,2,3) + k(4,-1,2)$
$l_2 : (x,y,z) = (2,-1,4) + m(1,0,-3)$

Method 1

Cross Product

$$\begin{array}{cccccc} 4 & -1 & 2 & 4 & -1 & 2 \\ 1 & 0 & -3 & 1 & 0 & -3 \end{array}$$

$= [3, 14, 1]$

$\therefore$ VE: $\vec{r} = [3,0,2] + t[3,14,1]$, $t \in \mathbb{R}$

Method 2

Let $[a,b,c]$ be the direction vector for the line $\perp$ to l_1 and l_2.

$[a,b,c] \cdot [4,-1,2] = 0$

$\boxed{4a - b + 2c = 0}$ ①

$[a,b,c] \cdot [1,0,-3] = 0$

$\boxed{a - 3c = 0}$ ②

Sub ② into ①

$4(3c) - b + 2c = 0$

$12c - b + 2c = 0$

$b = 14c$

let $c = 1$, $\Rightarrow a = 3$, $b = 14$

$\therefore \vec{r} = [3,0,2] + t[3,14,1]$, $t \in \mathbb{R}$.

b) parallel to the line $(x,y,z) = (-2,6,7) + t(-2,1,3)$

$\vec{r} = [3,0,2] + t[-2,1,3]$, $t \in \mathbb{R}$.

2. Find three points on the line $(x,y,z)=(-5,3,7)+s(2,-1,4)$

Let $s=0 \Rightarrow (-5,3,7)$

Let $s=1 \Rightarrow (-3,2,11)$

Let $s=2 \Rightarrow (-1,1,15)$

3. Are $\frac{x-4}{2}=y-6=z-5$ and $(x,y,z)=(6,8,7)+k(-4,-2,-2)$ the same line?

$\vec{d_1}=[2,1,1]$; $\vec{d_2}=[-4,-2,-2]$

$\therefore \vec{d_2}=-2\vec{d_1} \Rightarrow$ the lines are parallel.

Find k: $(4,6,5)=(6,8,7)+k(-4,-2,-2)$

$4=6-4k \rightarrow k=\frac{1}{2}$

$6=8-2k \rightarrow k=1$

$5=7-2k \rightarrow k=1$

Since k values are <u>not</u> the same, the lines are <u>not</u> the same (coincident).